

Keywords: Age at trauma exposure, Trauma type, Fear conditioning, Fear Extinction, fMRI

922. Reward Anticipation in Early Expression of Psychotic Disorder: A Functional MRI Approach

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Background: Previous research in patients with psychotic disorder has shown reduced activation in the brain's reward system (ventral medial prefrontal cortex (VMPFC), ventral striatum (VS), nucleus accumbens, caudate, putamen, ventral tegmental area (VTA)). The literature on the reward system in subclinical psychosis is limited and inconclusive, describing both hypo- and hyper-activation. Therefore, reward anticipation in individuals with subclinical psychotic symptoms was further examined.

Methods: A sample of young adults aged 16-26 years (n=25) with subclinical psychosis and healthy controls (n=42) underwent fMRI scanning. The Monetary Incentive Delay task was conducted with cues related to win, loss or neutral conditions with three gain or loss levels (small; €0.10, medium; €0.60 and large €3.00). fMRI analysis was conducted using FSL's General Linear Model on the reward versus neutral contrast. Whole brain analysis and region of interest (ROI) analysis on the VMPFC, VS and VTA was conducted. After extracting ROI data, statistical analysis was performed by regression analysis in STATA.

Results: Main effects of the large win (€3.00) > neutral contrast were found in both groups showing widespread brain activation. The PE > HC group comparison on the large win > neutral contrast showed significant increased activation in the left temporal pole. The ROI analysis revealed equal activation during reward anticipation between both groups.

Conclusions: The increased activation during reward anticipation in individuals with subclinical psychosis may point to mechanisms of plasticity and/or compensation. Reward related regions were not different during the anticipation phase, and suggests that the underlying dopamine dysregulation may be subtle in this phase.

Keywords: Psychotic Disorders, fMRI, reward anticipation, Monetary Gain, Prodrome

923. Maternal Caregiving Moderates Brain-Emotional Behavior Relationships in 3 Month Infants

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Background: The relationship between typical development of infant emotionality and brain measures, and moreover, how

maternal caregiving might influence this relationship remains unclear. Finding biological correlates of temperament during infancy would not only better our understanding of brain-behaviour relationships, but may be helpful in predicting future psychological outcome empirically. Our aims were to (1) investigate the relationship between diffusion metrics and 3- and 9-month emotionality measures, and (2) examine whether this relationship is moderated by maternal caregiving.

Methods: Mother-infant dyads were recruited through the Pittsburgh Girls Study. Diffusion tensor images were collected at 3 months and mother-infant interactions were collected at 3 and 9 months. Deterministic tractography was performed using ExploreDTI. General linear models were developed to examine the relationship between diffusion measures, including fractional anisotropy (FA), longitudinal and radial diffusivity, and infant temperament. Next, maternal caregiving measures were tested for moderating effects.

Results: Here we found a (1) positive relationship between FA of the right uncinate fasciculus and 3 month negative to positive emotionality ratio [$F(1,22)=5.57$, $p=0.028$], (2) positive association between FA of the left uncinate fasciculus and 9 month self soothability [$F(1,12)=16.5$, $p=0.0016$], and (3) moderation effect of maternal sensitivity on relationship (1) [$F(3,20)=3.83$, $p=0.016$].

Conclusions: Our results provide evidence for a relationship between diffusivity in the uncinate fasciculus and infant temperament measures at 3 and 9 months. Moreover, this relationship appears to be enhanced with greater maternal sensitivity. Our results suggest the importance of emotional experience and maternal care for the development and refining of emotional circuits.

Supported By: NIH R21

Keywords: Diffusion Tensor Imaging (DTI), Infant Temperament, Maternal sensitivity, Predictive Analytics, structural neuroimaging

924. Socioeconomic Status in Early Childhood Predicts White Matter Integrity in Young Adulthood

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Background: Socioeconomic disadvantage confers higher risk for medical illness, psychiatric distress, and poor cognitive functioning (Gianaros, 2013). Changes in brain structure may be one mechanism linking low socioeconomic status (SES) to poor health outcomes (Ursache & Noble, 2016). Concurrent associations between SES and white matter microstructure have been reported in childhood (Ursache & Noble, 2016), late adolescence (Noble et al., 2013), and adulthood (Gianaros et al., 2013) in the cingulum, a pathway implicated in executive functioning and self-regulation. Longitudinal investigations are necessary to determine whether SES in early life can impact white matter microstructure later in development.

Methods: High-risk male participants (N=158) were recruited in infancy based on low family income and followed until young adulthood as part of the Pitt Mother & Child Project.

SES was estimated using a composite score derived from family income and neighborhood impoverishment at ages 1.5, 2, and 3.5. Subjects underwent diffusion tensor imaging (61 directions, $b=1000$) on a 3T Siemens TIM Trio at age 20. Preprocessing was conducted in FSL, Tract-Based Spatial Statistics (TBSS) was used to calculate measures of white matter microstructure, and mean fractional anisotropy (FA) was calculated for the left and right cingulum.

Results: Lower SES in early childhood was associated with significantly lower FA in early adulthood in the right ($\beta=.188$, $p=.019$) and left ($\beta=.177$, $p=.028$) cingulum.

Conclusions: Early socioeconomic disadvantage is linked to lower cingulum FA in young adulthood. Enduring impairments in cingulum microstructure may confer increased risk for poor physical, psychological, and cognitive functioning across development.

Supported By: NIH R01 DA026222

Keywords: socioeconomic status, diffusion MRI, longitudinal, Neurodevelopment

925. Global Probabilistic Tractography and Symptom Dimensions in a Prospectively Characterized Sample of Adults with a Childhood Diagnosis of Attention Deficit-Hyperactivity Disorder

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Background: White matter abnormalities have been shown to play an important role in the pathophysiology of Attention Deficit-Hyperactivity Disorder (ADHD). However, little is known about the extent to which ADHD symptom severity is associated with abnormalities in white matter tracts known to be involved in attention and emotional control processes.

Methods: We aimed to determine if abnormalities in fronto-temporal white matter tracts involved in attention (superior longitudinal fasciculus, SLF) and emotional control processes (uncinate fasciculus, UF) relate to clinically relevant symptom-dimensions in 46 adults with or without ADHD (32 ADHD, 14 non-ADHD; mean age[SD]=33[3] years, 44males), recruited from an ongoing longitudinal study in 347 children with ADHD prospectively characterized from-youth-to-adulthood. Symptom-dimensions of inattention, hyperactivity/impulsivity (H/I), and anger-irritability (A/I) were used. Global probabilistic tractography was used to reconstruct SLF and UF. Volume, length, and diffusivity metrics were extracted for each participant.

Results: ADHD, vs non-ADHD, adults showed smaller volume in the right UF ($p=.05$) and right SLF ($p=.06$). In ADHD adults, H/I symptoms were negatively correlated with length in the SLF (left: $r=-.40$, $p=0.01$; right: $r=-.40$, $p=0.03$) and A/I symptoms were positively correlated with volume of the right UF ($r=0.40$, $p=0.04$).

Conclusions: Findings suggest that higher volumes in fibers connecting medial-temporal and orbitofrontal regions might be associated with higher severity of A/I symptoms. Abnormal reorganization of the fibers connecting DLPFC and

temporo-parietal regions, as evidenced by a shorter length in the SLF, may represent a neurobiological substrate of higher levels of H/I symptoms, possibly associated with inability of modulating thoughts and actions in goal-directed behaviors reported in ADHD.

Supported By: NIMH (R01MH101096-03)

Keywords: global probabilistic tractography, Symptom Dimensions, Adults with a Childhood Diagnosis of Attention Deficit-Hyperactivity Disorder, hyperactivity/impulsivity, anger-irritability

926. Regional Differences and Demographic Correlates of Cerebrovascular Reactivity Among Healthy Adolescents

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Background: Cerebrovascular reactivity (CVR), a proxy of cerebrovascular health, is reduced in adult depression. Breath-holding (BH) paradigms alter vasoactive CO₂, are well tolerated, and produce robust CVR measures in adults. We therefore look to extend BH-CVR to a healthy adolescents (HA) sample, investigating two modelling approaches, evaluate regional differences in BH-CVR, and examine BH-CVR in relation to demographics (e.g. sex).

Methods: Thirty-nine HA (ages 13-20 years, 20 females) completed six 15-second BHs, alternating with 30-second free-breathing intervals. Blood-oxygenation-level dependent (BOLD) fMRI at 3-Tesla measured CVR voxel-wise and in five major brain subdivisions. Hemodynamic responses were modelled using a: 1) double-gamma variate convolved with a boxcar function with an individualized delay term (HDR-Delay), and 2) a sine-cosine regressor with a delay term. CVR-delay was the elapsed time between end of BH and peak BOLD signal.

Results: There were regional differences in CVR (Frontal >Occipital >Parietal >Temporal >Subcortical) and CVR-delay (Occipital >Parietal >Frontal >Temporal >Subcortical). Males had higher CVR than females ($p=.03$, partial $\eta^2=.35$). Comparing both models, the HDR-Delay method yielded a significantly superior model fit and marginally larger cluster volume than the sine-cosine regressor.

Conclusions: This study found regional CVR differences in HA, which can inform future studies of adolescents with brain- and/or vascular-related diseases. Our regional CVR hierarchy findings were consistent with prior healthy adult studies, except for the relatively higher frontal CVR. Lower CVR in females converges with prior evidence of higher basal cerebral blood flow. Both methods for modelling the raw CVR data were effective, albeit the HDR-Delay method was marginally superior.

Supported By: Ontario Mental Health Foundation

Keywords: Cerebrovascular Reactivity, Breath-Hold, Magnetic resonance imaging, Healthy subjects, Adolescents